

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION CONTROL, LOW-VISCOSITY GRADES, AND DOWNSTREAM PROCESSING

1. ABSTRACT

This report synthesizes a set of patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-rubber technologies. The target is defined here as NBR grades with reduced acrylonitrile content or compositions/processes that enable low-ACN performance through monomer feed control, molecular-weight reduction, or formulation strategies that improve processability and end-use properties. Across the selected patents, the landscape spans three major technical directions: (1) direct synthesis or modification of nitrile rubber to lower molecular weight and viscosity, including olefin-metathesis degradation followed by hydrogenation; (2) composition engineering of highly saturated or carboxylated nitrile rubbers with polyamide, silica, fillers, and cross-linking systems to balance oil resistance, heat resistance, rollability, and compression set; and (3) downstream compounding or masterbatch approaches that improve dispersion of cross-linkers or additives and suppress foaming or viscosity rise during processing.

The most synthesis-relevant disclosures are the metathesis-based patents, which show that nitrile rubber can be reacted in solution with ruthenium or related metathesis catalysts, often in chlorinated aromatic solvents, to reduce molecular weight and narrow dispersity before optional hydrogenation. These routes are directly relevant to low-ACN NBR because they address processability limits associated with high Mooney viscosity while preserving nitrile-rubber chemistry. Other patents focus on highly saturated nitrile rubbers containing 15–60 wt% acrylonitrile units, often with carboxylated comonomers such as mono-n-butyl maleate, and demonstrate that blending with polyamide resin and using peroxide or amine-based cross-linking systems can improve rollability and high-temperature strength. Several patents also emphasize controlled compounding conditions, including twin-screw extrusion at 220–320°C, Banbury mixing, and masterbatch preparation at or above the melting point of the amine cross-linker.

Overall, the evidence suggests that the most useful synthesis window for low-acrylonitrile NBR development is not a single monomer recipe but a combination of composition control, low-viscosity generation, and formulation optimization. The strongest recurring trends are reduced molecular weight, controlled acrylonitrile content, selective hydrogenation, and carefully tuned cross-linking/dispersion strategies to preserve oil resistance while improving processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis degradation of nitrile rubber in solution using a transition-metal metathesis catalyst, followed by hydrogenation to obtain hydrogenated nitrile rubber with reduced molecular weight. The examples show dissolution of nitrile rubber in monochlorobenzene, addition of 1-hexene and a ruthenium metathesis catalyst, then hydrogenation with a rhodium catalyst under hydrogen pressure. - Nitrile rubber charge: 700 g — Examples 5-6 - Solvent charge: 4667 g monochlorobenzene — Examples 5-6 - Reactor volume: 10 L — Examples 5-6 - 1-Hexene addition: 4 phr — Examples 5-6 - Metathesis reaction temperature: 22°C — Examples 5-6 - Metathesis reaction time: 2 h — Examples 5-6 - Hydrogenation catalyst loading: 0.06 phr tris-(triphenylphosphine) rhodium chloride — Examples 5-6 - Hydrogen pressure: 85 bar — Examples 5-6 - Hydrogenation temperature: 138°C — Examples 5-6 - Hydrogenation time: 4 h — Examples 5-6 - Agitation speed: 600 rpm — Examples 5-6 - Metathesis catalyst loading: 0.1 phr — Example 7 - Solvent content: 1,300 ppm — Example 7 - Mn: 19,000 g/mol — Example 7 - Mw: 34,000 g/mol — Example 7 - PDI: 1.78 — Example 7 - Control Mn: 69,000 g/mol — Example 7 - Control Mw: 217,500 g/mol — Example 7 - Control PDI: 3.15 — Example 7

Relevant Experimental Evidence - Example 1: metathesis catalyst loading of 0.04 phr gave reduced Mn and Mw relative to control. - Example 2: increasing catalyst loading to 0.06 phr further lowered molecular weight. - Example 3: 0.08 phr catalyst continued the molecular-weight reduction trend. - Example 4: 0.1 phr catalyst produced the lowest Mn/Mw among Examples 1-4. - Example 7: changing the nitrile rubber feedstock still gave low-Mw HNBR after metathesis and hydrogenation.

Technical Relevance This is highly relevant because it provides a direct route to low-viscosity nitrile rubber by metathesis degradation, which is a practical way to obtain low-Mooney or low-molecular-weight NBR/HNBR feedstocks. Although not a low-acrylonitrile synthesis per se, it addresses the processability problem that often motivates low-ACN grade development.

Patent 2

- Patent Number: EP3323850B1
- Patent Title: Nitrile rubber composition and cross-linked rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a compounding and cross-linking disclosure for nitrile rubber compositions rather than a monomer-synthesis patent. It prepares nitrile rubber compositions using carboxyl group-containing highly saturated nitrile rubber, low-surface-area silica, polyamine cross-linking agents, trimellitic acid plasticizers, and optional silane coupling agents, then evaluates cured properties. - Silica specific surface area: 48 m²/g — Example 1 - Silica amount: 50 parts — Example 1 - Tri-2-ethylhexyl trimellitate: 5 parts — Example 1 - 4,4'-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - Polyoxyethylene alkyl ether phosphoric acid ester: 1 part — Example 1 - DBU: 4 parts — Example 1 - Hexamethylenediamine carbamate: 2.4 parts — Example 1 - Mixing temperature: 50°C — Example 1 - Mixing time: 5 minutes — Example 1 - Silane coupling agent 3-aminopropyltriethoxysilane: 1 part — Example 2 - Silica specific surface area: 45 m²/g — Examples 3-4 - Silica amount: 55 parts — Example 5 - Hexamethylenediamine carbamate: 2.7 parts — Example 5 - Silica amount: 60 parts — Example 6 - Polyether ester: 5 parts — Example 6 - Hexamethylenediamine carbamate: 1.9 parts — Example 6 - Silica specific surface area: 55 m²/g — Example 7 - Silane coupling agent 3-aminopropyltriethoxysilane: 1 part — Example 8 - Carboxyl group-containing highly saturated nitrile rubber (a1): 100 parts — Example 1 - Carboxyl group-containing highly saturated nitrile rubber (a2): 100 parts — Example 5 - Carboxyl group-containing highly saturated nitrile rubber (a3): 100 parts — Example 6 - Iodine value: 120 or less — Claims - Nitrile monomer unit content: 5 to 60 wt% — Claims - Silica content: 10 to 100 parts by weight — Claims - Polyamine cross-linking agent content: 0.1 to 20 parts by weight — Claims

Relevant Experimental Evidence - Example 1: low-surface-area silica, trimellitate plasticizer, DBU, and hexamethylenediamine carbamate produced a nitrile rubber composition with good physical properties. - Example 2: adding 3-aminopropyltriethoxysilane after silica addition was used to assess silane timing effects. - Example 5: increasing silica and cross-linker levels with a different highly saturated nitrile rubber variant was evaluated. - Example 6: replacing trimellitate with polyether ester and reducing diamine level showed formulation flexibility. - Example 8: using a different silica and silane coupling agent again demonstrated formulation robustness.

Technical Relevance This patent is relevant because it defines formulation windows for highly saturated nitrile rubber systems that can include lower-acrylonitrile nitrile rubber feedstocks. It is especially useful for downstream compounding of low-ACN NBR where silica, plasticizer, and

polyamine cross-linking must be balanced for processability and cured performance.

Patent 3

- Patent Number: EP2671914B1
- Patent Title: Nitrile rubber composition, crosslinkable nitrile rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubber compositions containing a nitrile rubber with 20 to 80 wt% acrylonitrile units, conjugated diene units, and optional cationic monomer units, combined with a layered inorganic filler having a controlled surface-area particle size distribution. The examples focus on preparing layered inorganic filler aqueous dispersions, blending them into nitrile rubber latex, coagulating, drying, and then evaluating the resulting cross-linked rubber. - Layered inorganic filler: 100 parts — Example 1 - Distilled water: 1898 parts — Example 1 - Sodium polyacrylate: 2 parts — Example 1 - Agitation speed: 7,000 rpm — Example 1 - Agitation temperature: 30°C — Example 1 - Agitation time: 30 minutes — Example 1 - Layered inorganic filler addition: 20 parts per 100 parts rubber solid — Example 1 - Plasticizer amount: 10 parts — Example 1 - Calcium chloride coagulant: 4 parts by weight per 100 parts rubber — Example 1 - Drying temperature: 60°C — Example 1 - Reduced pressure drying — Example 1 - Acrylonitrile units: 50 wt% — Example 5 - Butadiene units: 48 wt% — Example 5 - 2-vinylpyridine units: 2 wt% — Example 5 - Plasticizer: 30 parts — Example 5 - Vinyl chloride resin: 66 parts — Example 5 - Acrylic resin: 66 parts — Example 5 - Iodine value: 23 — Example 6 - Acrylonitrile units: 50 wt% — Example 6 - Butadiene units (including hydrogenated parts): 48 wt% — Example 6 - 2-vinylpyridine units: 2 wt% — Example 6 - Plasticizer: 30 parts — Example 6 - Surface area particle size distribution d50: 0.20 µm to 6.0 µm — Claims - Surface area particle size distribution d90-d10: 1.1 µm or more — Claims - Layered inorganic filler content: 1 to 100 parts by weight — Claims - Mooney viscosity: 3 to 250 — Claims - Aspect ratio: 30 to 2000 — Claims - Cationic monomer content: 0 to 30 wt% — Claims

Relevant Experimental Evidence - Example 1: a layered inorganic filler dispersion was prepared at 7,000 rpm and 30°C, then coagulated with calcium chloride to form nitrile rubber composition. - Example 2: adding vinyl chloride resin to the latex composition was used to assess thermoplastic resin effects. - Example 5: replacing vinyl chloride resin with acrylic resin showed formulation flexibility while maintaining performance. - Example 6: using hydrogenated nitrile rubber (iodine value 23) demonstrated applicability to HNBR-type feedstocks. - Comparative examples: poor dispersion or overly strong dispersion degraded gasoline permeation resistance or cold resistance.

Technical Relevance This patent is relevant because it addresses nitrile-rubber composition design around acrylonitrile-containing backbones and shows how filler dispersion and latex

processing affect final properties. It is useful for low-ACN NBR development when balancing oil resistance, cold resistance, and processability in filled compounds.

Patent 4

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis-based segment interchange between two or more ethylenically unsaturated polymers to form meta-block copolymers. The examples use partially hydrogenated nitrile/butadiene rubber with other unsaturated polymers in solution, with Grubbs II or tungsten-based metathesis catalysts, followed by precipitation and recovery of the metathesized product. - Polymer charge: 0.25 g each HPBD and HNBR — Example 1 - Solvent: toluene — Example 1 - Dissolution temperature: 95°C — Example 1 - Catalyst charge: 8 mg Grubbs II metathesis catalyst — Example 1 - Reaction time: 1 hour — Example 1 - Precipitation solvent: methanol — Example 1 - Polymer charge: 0.25 g each unsaturated polyurethane and poly(ethylene-co-butadiene) — Example 2 - Dissolution temperature: 105°C — Example 2 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 2 - THF extraction: 30 ml — Example 2 - Polymer charge: 1.05 g fumaryl-modified polycarbonate and 0.45 g poly(ethylene-co-butadiene) — Example 3 - Dissolution temperature: 105°C — Example 3 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 3 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-butadiene) — Example 4 - Dissolution temperature: 105°C — Example 4 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 4 - Polymer charge: 0.25 g each unsaturated poly(ethylene oxide) and poly(ethylene-co-butadiene) — Example 5 - Dissolution temperature: 105°C — Example 5 - Catalyst charge: 9 mg Grubbs II catalyst — Example 5 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) — Example 6 - Dissolution temperature: 105°C — Example 6 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 6 - Polymer charge: 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) — Example 7 - Dissolution temperature: 105°C — Example 7 - Tungsten hexachloride catalyst: 8,000 ppm — Example 7 - Tri-n-butylmethyltin co-catalyst: 13,000 ppm — Example 7 - Polymer charge: 0.25 g each unsaturated polyester and poly(ethylene-co-butadiene) — Example 8 - Dissolution temperature: 105°C — Example 8 - Catalyst charge: 9 mg Grubbs II catalyst — Example 8 - Polymer charge: 0.25 g SBS and 0.25 g poly(ethylene-co-butadiene) — Example 9 - Dissolution temperature: 105°C — Example 9 - Tungsten hexachloride catalyst: 8,000 ppm — Example 9 - Tri-n-butylmethyltin co-catalyst: 13,000 ppm — Example 9 - Polymer charge: 0.25 g unsaturated fumaryl-modified polycarbonate, 0.25 g

SBS, and 0.25 g poly(ethylene-co-butadiene) — Example 10 - Dissolution temperature: 105°C — Example 10 - Catalyst charge: 9 mg Grubbs II catalyst — Example 10 - THF extraction: 50 ml — Example 10 - Unsaturation in at least one reagent: 0.001 to 10 mole percent — Claims

Relevant Experimental Evidence - Example 1: HNBR and HPBD were metathesized to form a meta-block copolymer with microphase separation. - Example 2: polyurethane and poly(ethylene-co-butadiene) underwent segment interchange confirmed by FTIR. - Example 3: fumaryl-modified polycarbonate and poly(ethylene-co-butadiene) formed an ordered meta-block copolymer. - Example 7: tungsten catalyst/co-catalyst system enabled segment interchange in polyolefin blends. - Example 10: a terpolymer-like meta-block product was formed from polycarbonate, SBS, and poly(ethylene-co-butadiene).

Technical Relevance This patent is relevant because it explicitly includes partially hydrogenated nitrile/butadiene rubber as a metathesis reagent, showing that nitrile-rubber segments can be reorganized into blocky architectures. While not a low-ACN synthesis route, it is important for understanding post-polymerization modification of nitrile rubber feedstocks and their processability.

Patent 5

- Patent Number: EP2561011B1
- Patent Title: Thermoplastic and/or elastomeric composite material based on carbon nanotubes ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses agglomerated solid composites of carbon nanotubes and graphenes in a thermoplastic or elastomeric polymer matrix. One example specifically uses a butadiene-acrylonitrile copolymer resin as the matrix, processed in a BUSS co-kneader with controlled temperature, flow rate, and screw speed to make granules that can later be diluted into a polymer matrix. - CNT content: 35 wt% — Example 7 - Graphene content: 5 wt% — Example 7 - Nitrile resin content: 55 wt% — Example 7 - Acrylic copolymer content: 5 wt% — Example 7 - Nipol 1312V preheat: 160°C — Example 7 - Injection temperature: 190°C — Example 7 - Co-kneader temperature: 200°C — Example 7 - Flow rate: 12 kg/h — Example 7 - Screw speed: 240 rpm — Example 7 - Granule drying temperature: about 50°C — Example 7 - CNT content: 30 wt% — Example 8 - Graphene content: 5 wt% — Example 8 - Silicone resin content: 65 wt% — Example 8 - Kneading temperature: 90-110°C — Example 8 - CNT content: 30 wt% — Example 9 - Graphene content: 5 wt% — Example 9 - Kneading temperature: 160-180°C — Example 9 - CNT content: 35 wt% — Example 10 - Graphene content: 5 wt% — Example 10 - Liquid fluoroelastomer resin content: 20 wt% — Example 10 - Kneading

temperature: 160-180°C — Example 10 - CNT content: 18 wt% — Example 11 - Graphene content: 2 wt% — Example 11 - Zone 1 temperature: 150°C — Example 11 - Zone 2 temperature: 140°C — Example 11 - Flow rate: 12 kg/h — Example 11 - Screw speed: 200 rpm — Example 11 - Claimed CNT content: 0.1 to 50 wt% — Claims - Claimed graphene content: 0.1 to 20 wt% — Claims

Relevant Experimental Evidence - Example 7: a nitrile-resin-based composite containing 35 wt% CNT and 5 wt% graphene was produced as granules. - Example 8: a silicone-elastomer-based composite was made at 90-110°C and granulated for later dilution. - Example 9: a fluorinated elastomer composite was prepared at 160-180°C with 30 wt% CNT and 5 wt% graphene. - Example 10: a fluorinated elastomer/liquid resin blend was processed into granules and suggested for gasoline transport pipes. - Example 11: staged feeding of CNT/graphene into a solid VITON feed showed controlled processing at 150/140°C.

Technical Relevance This patent is relevant because it explicitly uses a butadiene-acrylonitrile copolymer resin as a matrix in a high-filler composite, showing compatibility of nitrile-based materials with CNT/graphene masterbatch processing. It does not synthesize low-ACN NBR, but it is useful for downstream compounding and application development of nitrile-rubber-based materials.

Patent 6

- Patent Number: US9306208B2
- Patent Title: Manufacturing method of an electrode for electrochemical element and an ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses manufacturing electrodes for electrochemical elements by compression-forming mixed powders or composite particles that include an alkaline metal or alkaline earth metal powder having a coated surface. The examples prepare coated lithium powder, mix it with electrode active material and binders, then form and heat-treat electrodes under controlled dry-room conditions. - Lithium dispersion in oil: 55.72 g including 6.28 g lithium — Example 1 - Heating temperature: 75°C — Example 1 - Styrene-butadiene copolymer solution: 1.9 g of 10% p-xylene solution — Example 1 - Stirring time: 22 hours — Example 1 - Stirring speed: 200 rpm — Example 1 - Coated film thickness: 20 nm — Example 1 - Mean volume particle diameter: 30 µm — Example 1 - Coat polymer amount: 3 parts per 100 parts lithium — Example 1 - Slurry solid content: 20% — Example 1 - Spray dryer air temperature: 150°C — Example 1 - Particle collecting outlet temperature: 90°C — Example 1 - Vacuum drying: 24 hours at 150°C — Example 1 - Moisture amount: 50 ppm or less — Example 1 - Dry mixing time: 10 minutes — Example 1 - Dry mixing speed: 500 rpm — Example 1 - Dry-room

dew point: -50°C or less — Example 1 - Coated lithium powder amount: 10 parts per 100 parts active material — Example 1 - Hard carbon particle diameter: $10\text{ }\mu\text{m}$ — Example 6 - Polycarbonate electrolyte concentration: 1 mole/L — Example 7 - Coated film thickness: 30 nm — Example 8 - Coated film thickness: 100 nm — Example 9 - CO_2 coating thickness: 5 nm — Example 10 - Wax coating thickness: 5 nm — Example 11 - Vacuum-drying condition: 100°C for 10 hours — Example 12 - Dry-room dew point: -30°C — Example 13 - Dry mixing time: 5 minutes — Example 14

Relevant Experimental Evidence - Example 1: coated lithium powder and spray-dried composite particles enabled a dry-mixed negative electrode. - Example 2: hard carbon was substituted for Si-O-C while keeping the coated lithium approach. - Example 3: an acrylate polymer binder was used in composite particles. - Example 4: higher lithium powder size reduction to $10\text{ }\mu\text{m}$ was achieved by changing stirring conditions. - Example 10: CO_2 -coated lithium powder was used as an alternative surface treatment. - Example 12: insufficient vacuum drying increased composite-particle moisture to 200 ppm.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it concerns electrode manufacturing rather than nitrile-rubber synthesis. It is included because it demonstrates the use of coated polymeric materials and dry mixing in a process environment where nitrile-based binders or elastomers may be relevant in battery applications.

Patent 7

- Patent Number: US10280298B2
- Patent Title: Highly saturated nitrile rubber composition and cross-linked rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses highly saturated nitrile rubber compositions made by blending a carboxyl group-containing highly saturated nitrile rubber with a second highly saturated nitrile rubber and a polyamide resin, then kneading at elevated temperature and adding peroxide cross-linker. The synthesis examples show emulsion polymerization of acrylonitrile/butadiene with maleate or methacrylic acid comonomers, followed by hydrogenation and compounding with Nylon 12 or Nylon 6. - Ion-exchanged water: 180 parts — Synthesis Example 1 - Sodium dodecylbenzene sulfonate solution: 25 parts of 10 wt% solution — Synthesis Example 1 - Acrylonitrile: 37 parts — Synthesis Example 1 - Mono n-butyl maleate: 6 parts — Synthesis Example 1 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-Butadiene: 57 parts — Synthesis Example 1 - Polymerization temperature: 5°C — Synthesis Example 1 - Cumen hydroperoxide: 0.1 part — Synthesis Example 1 - Polymerization time: 16 hours — Synthesis Example 1 - Hydrogenation pressure: 3 MPa — Synthesis Example 1 -

Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Methanol coagulation: 2 volumes — Synthesis Example 1 - Vacuum drying: 60°C for 12 hours — Synthesis Example 1 - Acrylonitrile units: 35.6 wt% — Synthesis Example 1 - Butadiene units: 59.0 wt% — Synthesis Example 1 - Mono n-butyl maleate units: 5.4 wt% — Synthesis Example 1 - Iodine value: 7 — Synthesis Example 1 - Mooney viscosity: 55 — Synthesis Example 1 - Ion-exchanged water: 200 parts — Synthesis Example 2 - Sodium carbonate: 0.2 part — Synthesis Example 2 - Fatty acid potassium soap: 2.25 parts — Synthesis Example 2 - Acrylonitrile: 42 parts — Synthesis Example 2 - t-Dodecyl mercaptan: 0.45 part — Synthesis Example 2 - 1,3-Butadiene: 58 parts — Synthesis Example 2 - Polymerization temperature: 5°C — Synthesis Example 2 - Polymerization time: 16 hours — Synthesis Example 2 - Hydrogenation pressure: 3.0 MPa — Synthesis Example 2 - Acrylonitrile units: 40.5 wt% — Synthesis Example 2 - Iodine value: 7 — Synthesis Example 2 - Mooney viscosity: 100 — Synthesis Example 2 - Ion-exchanged water: 180 parts — Synthesis Example 3 - Sodium dodecylbenzene sulfonate solution: 25 parts of 10 wt% solution — Synthesis Example 3 - Acrylonitrile: 37 parts — Synthesis Example 3 - Methacrylic acid: 4 parts — Synthesis Example 3 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 3 - 1,3-Butadiene: 59 parts — Synthesis Example 3 - Polymerization temperature: 5°C — Synthesis Example 3 - Polymerization time: 16 hours — Synthesis Example 3 - Hydrogenation carried out as in Synthesis Example 1 — Synthesis Example 3 - Acrylonitrile units: 36 wt% — Synthesis Example 3 - Butadiene units: 61 wt% — Synthesis Example 3 - Methacrylic acid units: 3 wt% — Synthesis Example 3 - Iodine value: 9 — Synthesis Example 3 - Mooney viscosity: 68 — Synthesis Example 3 - Carboxyl group-containing highly saturated nitrile rubber (a1): 21 parts — Example 1 - Highly saturated nitrile rubber (a2): 49 parts — Example 1 - Nylon 12: 30 parts — Example 1 - Twin screw extruder temperature: 220°C — Example 1 - MT carbon black: 40 parts — Example 1 - Tri-2-ethylhexyl trimellitate: 10 parts — Example 1 - 4,4'-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - 1,3-bis(t-butylperoxyisopropyl)benzene 40% product: 7 parts — Example 1 - Twin screw extruder temperature: 320°C — Example 2 - Nylon 6: 30 parts — Example 3 - Carboxyl group-containing highly saturated nitrile rubber (a1): 10.5 parts — Example 4 - Highly saturated nitrile rubber (a2): 59.5 parts — Example 4 - Carboxyl group-containing highly saturated nitrile rubber (a1): 24 parts — Example 5 - Highly saturated nitrile rubber (a2): 56 parts — Example 5 - Nylon 12: 20 parts — Example 5

Relevant Experimental Evidence - Synthesis Example 1: carboxylated NBR was emulsion-polymerized at 5°C, hydrogenated at 3 MPa and 50°C, and gave a low-iodine highly saturated nitrile rubber. - Synthesis Example 2: a non-carboxylated highly saturated nitrile rubber with 40.5 wt% acrylonitrile and Mooney 100 was prepared. - Synthesis Example 3: methacrylic-acid-containing highly saturated nitrile rubber was prepared and hydrogenated. - Example 1: blending carboxylated and non-carboxylated HNBR with Nylon 12 at 220°C gave a cross-linkable composition. - Example 2: raising the twin-screw kneading temperature to 320°C was used to compare rollability and properties. - Example 3: replacing Nylon 12 with Nylon 6 evaluated polyamide selection effects.

Technical Relevance This patent is highly relevant because it directly discloses nitrile-rubber synthesis with acrylonitrile contents in the low-to-mid range and shows how carboxylated and

non-carboxylated HNBR components can be combined for improved rollability and high-temperature performance. It provides explicit monomer ratios and hydrogenation conditions useful for designing low-ACN nitrile rubber feedstocks.

Patent 8

- Patent Number: US9382391B2
- Patent Title: Cross-linkable nitrile rubber composition and method of production of same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a masterbatch approach for cross-linkable nitrile rubber compositions. A nitrile copolymer rubber with very low carboxyl content is first combined with an aromatic polyvalent amine cross-linker at or above the cross-linker melting point to form a master batch, which is then blended into a carboxyl group-containing nitrile copolymer rubber to improve dispersibility and suppress foaming. - Hydrogenated nitrile rubber (Zetpol 2000L): 100 parts — Example 1 - Acrylonitrile unit content: 36 wt% — Example 1 - Butadiene units (including hydrogenated): 64 wt% — Example 1 - Iodine value: 4 — Example 1 - Mooney viscosity: 65 — Example 1 - Carboxyl group content: 5×10^{-6} ephr or less — Example 1 - Master batch temperature: 130°C — Example 1 - Melting point of 2,2-bis[4-(4-aminophenoxy)phenyl]propane: 128°C — Example 1 - 2,2-bis[4-(4-aminophenoxy)phenyl]propane: 100 parts — Example 1 - FEF carbon black: 40 parts — Example 1 - Trimellitic acid ester: 5 parts — Example 1 - Nauguard 445: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - Phosphanol RL210: 1 part — Example 1 - Master batch (B-1): 6.77 parts — Example 1 - Noccelar DT: 2 parts — Example 1 - Diak #1: 2.6 parts — Example 1 - Master batch (B-2): 10.2 parts — Example 2 - 2,2-bis[4-(4-aminophenoxy)phenyl]propane: 50 parts — Example 2 - Iodine value: 120 or less — Claims - Master batch content: 0.1 to 40 parts by weight — Claims - Carboxyl group-containing nitrile copolymer rubber (A): 100 parts basis — Claims

Relevant Experimental Evidence - Example 1: a master batch made above the amine melting point improved dispersibility and produced a good cross-linkable composition. - Example 2: lowering the amine content in the master batch still gave a workable composition. - Comparative Example 1: direct addition of aromatic amine without masterbatching gave poor dispersibility and foaming. - Comparative Example 2: direct addition of aliphatic diamine was used as a contrast to the aromatic amine masterbatch approach.

Technical Relevance This patent is relevant because it shows a practical compounding strategy for nitrile-rubber systems with controlled carboxyl functionality and low iodine value. While not a synthesis patent, it is important for low-ACN NBR processing because it improves cross-linker

dispersion and final mechanical performance.

Patent 9

- Patent Number: US8034881B2
- Patent Title: Cross-linkable nitrile rubber composition and cross-linked rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses cross-linkable nitrile rubber compositions based on highly saturated nitrile rubber, a polyamine-based cross-linking agent, and a cyclic amidine or cyclic amidinium salt. The examples focus on compounding with carbon black, plasticizer, stearic acid, and different amidine additives to improve heat resistance and compression set. - Highly saturated nitrile rubber (a1): 100 parts — Example 1 - FEF carbon black: 40 parts — Example 1 - Trimellitic acid ester: 5 parts — Example 1 - Stearic acid: 1 part — Example 1 - Mixed phosphorus acid ester: 1 part — Example 1 - 4,4'-di-(α,α' -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - 2-mercaptobenzimidazole: 1.5 parts — Example 1 - VUCOFAC ACT 55: 2 parts — Example 1 - Hexamethylene diamine carbamate: 2.6 parts — Example 1 - RHENOGRAN XLA-60: 4 parts — Example 2 - ALCANPOUDRE DBU 70-3KG: 1 part — Example 3 - Iodine value: 120 or less — Claims - Cyclic amidine/cyclic amidinium salt content: 0.1 to 20 parts by weight — Claims - Polyamine-based cross-linking agent content: 0.1 to 20 parts by weight — Claims - Carboxyl-group containing monomer unit: α,β -ethylenically unsaturated dicarboxylic acid monoester monomer unit — Claims

Relevant Experimental Evidence - Example 1: DBU-containing additive plus hexamethylene diamine carbamate produced a cross-linkable nitrile rubber composition. - Example 2: a different DBU formulation (RHENOGRAN XLA-60) was evaluated. - Example 3: a DBU filler formulation was tested as another cyclic amidine source. - Comparative Examples 1-2: replacing the cyclic amidine system with other accelerators worsened compression set performance.

Technical Relevance This patent is relevant because it targets low-compression-set, heat-resistant nitrile rubber formulations that can be built from highly saturated nitrile rubber feedstocks. It is useful for low-ACN NBR compounding where cure efficiency and long-term mechanical stability are important.

Patent 10

- Patent Number: US10317811B2
- Patent Title: Charging member, method for producing same, process cartridge and ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a charging roller made from a vulcanized rubber layer based on a polymer having a butadiene skeleton, with insulating particles exposed at the surface and the surface roughened by ultraviolet irradiation after vulcanization. The examples use acrylonitrile-butadiene rubber as the surface-layer polymer and evaluate surface hardness, viscosity, and image performance. - Acrylonitrile-butadiene rubber: 100 parts — Example 1 - Carbon black: 50 parts — Example 1 - Zinc oxide: 5 parts — Example 1 - Calcium carbonate: 30 parts — Example 1 - Zinc stearate: 1 part — Example 1 - Capsule particle 1: 5 parts — Example 1 - Sulfur: 1 part — Example 1 - Tetrabenzyl thiuram disulfide: 4 parts — Example 1 - Molding temperature: 100°C — Example 1 - Screw rotations: 9 rpm — Example 1 - Average take-up rate: 107% — Example 1 - Vulcanization temperature: 160°C — Example 1 - Vulcanization time: 1 hour — Example 1 - Ultraviolet wavelength: 254 nm — Example 1 - Integrated UV light amount: 9000 mJ/cm² — Example 1 - Grinding grain size: 100 mesh — Comparative Example 6 - Roller rotations: 400 rpm — Comparative Example 6 - Grinding stone rotations: 2500 rpm — Comparative Example 6 - Incision rate: 20 mm/min — Comparative Example 6 - Spark out time: 1 second — Comparative Example 6 - Grinding margin: 400 μm — Comparative Example 6 - Outer diameter difference: 200 μm — Comparative Example 6 - Claimed Martens' hardness: 2 to 20 N/mm² — Abstract - Claimed viscosity: 70×10⁻³ V or smaller — Abstract

Relevant Experimental Evidence - Example 1: ABNR-based roller with roughened UV-treated surface was prepared by crosshead extrusion and air vulcanization. - Comparative Example 5: omitting UV irradiation was used to show the importance of surface treatment. - Comparative Example 6: grinding before UV treatment was used to compare surface roughening methods. - Examples 11-20: different particle types and surface conditions were evaluated for image performance and durability.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it uses acrylonitrile-butadiene rubber as a functional elastomeric layer rather than as a synthesis target. It is still useful as an application example showing that NBR-based materials can be formulated for demanding surface-performance roles.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473279B1 and EP2057205B1 both use metathesis chemistry, but EP2473279B1 focuses on degrading nitrile rubber to lower molecular weight before hydrogenation, whereas

EP2057205B1 uses metathesis to interchange segments between polymers and form meta-block copolymers.

- US10280298B2 and US9382391B2 both rely on highly saturated nitrile rubber with carboxyl functionality, but US10280298B2 emphasizes polyamide blending and peroxide cure, while US9382391B2 emphasizes masterbatching an aromatic polyvalent amine cross-linker for better dispersibility.
- EP3323850B1 and US8034881B2 are both formulation patents for nitrile-rubber compounds, but EP3323850B1 centers on silica/plasticizer/polyamine systems for low heat generation, whereas US8034881B2 centers on cyclic amidine additives to improve compression set and heat resistance.
- EP2671914B1 is the most latex- and filler-dispersion-oriented patent, using layered inorganic fillers with controlled particle-size distributions, while the other composition patents rely more on carbon black, silica, or polyamide reinforcement.
- EP2561011B1 is not a nitrile-rubber synthesis patent, but it uniquely includes a butadiene-acrylonitrile copolymer as a matrix for CNT/graphene masterbatch processing, linking nitrile rubber to conductive composite manufacturing.
- US9306208B2 and US10317811B2 are application/process patents rather than NBR synthesis patents; US9306208B2 concerns electrochemical electrodes with coated lithium powders, while US10317811B2 concerns charging rollers using A-BNR surface layers and UV roughening.
- Across the synthesis-relevant patents, acrylonitrile content is typically controlled in the 15 to 60 wt% range for highly saturated nitrile rubbers, with low-viscosity/processability improvements achieved by metathesis, hydrogenation, or formulation architecture rather than by extreme changes in ACN content.
- Processing temperatures vary widely: metathesis/hydrogenation routes use 22°C to 138°C in solution, while compounding routes use 130°C to 320°C, and twin-screw extrusion commonly appears at 220°C or higher for polyamide-containing systems.

4. CONCLUSION

The patent set shows that low-acrylonitrile NBR development is best approached as a combined composition-and-process problem rather than a single monomer specification. The strongest synthesis evidence comes from metathesis/hydrogenation routes that reduce molecular weight and improve processability, while the strongest formulation evidence comes from highly saturated nitrile rubber systems that use carboxylated and non-carboxylated components, polyamide reinforcement, and carefully selected cure packages. For practical low-ACN NBR synthesis, the evidence supports maintaining acrylonitrile in the lower end of the disclosed nitrile-rubber ranges, using controlled emulsion polymerization at low temperature, and then tailoring molecular weight by metathesis or related post-polymerization modification. For downstream processing, the most robust formulations use low-surface-area silica or polyamide reinforcement, with peroxide or amine-based cross-linking systems selected to match the intended cure route. Overall, the recommended evidence-based strategy is: low-temperature emulsion copolymerization of

butadiene/acrylonitrile with optional carboxyl comonomer, followed by selective hydrogenation or metathesis-based viscosity reduction, then compounding with a cure package optimized for rollability, compression set, and high-temperature strength.

5. REFERENCES

- EP2473279B1 | Metathesis of a nitrile rubber in the presence of a transition metal catalyst | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473279B1/en>
- EP3323850B1 | Nitrile rubber composition and cross-linked rubber | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3323850B1/en>
- EP2671914B1 | Nitrile rubber composition, crosslinkable nitrile rubber composition and ... | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2671914B1/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- EP2561011B1 | Thermoplastic and/or elastomeric composite material based on carbon nanotubes ... | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2561011B1/en>
- US9306208B2 | Manufacturing method of an electrode for electrochemical element and an ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9306208B2/en>
- US10280298B2 | Highly saturated nitrile rubber composition and cross-linked rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10280298B2/en>
- US9382391B2 | Cross-linkable nitrile rubber composition and method of production of same | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9382391B2/en>
- US8034881B2 | Cross-linkable nitrile rubber composition and cross-linked rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US8034881B2/en>
- US10317811B2 | Charging member, method for producing same, process cartridge and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10317811B2/en>